

Tournaments not inducible by five voters

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Abstract

A tournament T is k -inducible if there are k linear orders on its vertex set such that, for every arc $i \rightarrow j$ of T , a majority of the orders rank i above j . For each odd k some tournament is not k -inducible, and $N(k)$ denotes the least order at which one occurs. Only $N(3)$ is known exactly: $N(3) = 8$, with 96 of the 6,880 tournaments on 8 vertices not 3-inducible [7]. The value $N(5)$ is open. Bachmeier et al. [1] showed that *some* tournament on 41 vertices is not 5-inducible, by an argument that compares how many tournaments exist with how many five linear orders can produce; the smallest explicit such tournament known to them had about 6.0×10^8 vertices. Our companion paper [2] brought the range to $12 \leq N(5) \leq 38$ and exhibited an explicit tournament of order 43: the quadratic residue, or Paley, tournament P_{43} .

Results. We develop a bespoke search algorithm that improves both ends of the range,

$$13 \leq N(5) \leq 23.$$

The upper bound comes from proving that the Paley tournament on 23 vertices is not 5-inducible, the case [1] reported it could not decide, its solver having “not terminate[d] within a total of six weeks”. The lower bound comes from an exhaustive analysis at order 12. We obtain several further results on the same problem. Paley(27) and Paley(31) are also not 5-inducible. Paley(19) is 5-inducible, but is not 5-inducible with *unit margin*, that is, by a profile in which every arc is carried by exactly three voters against two. Paley(23) is arc-critical: reversing any of its arcs makes it 5-inducible. Paley(43), by contrast, is *not* vertex-critical: deleting a vertex leaves a 42-vertex tournament that is still not 5-inducible. Additionally, we verify that every self-converse tournament on at most 13 vertices and every regular tournament on at most 15 vertices is 5-inducible with unit margin.

Method. All of these results are proved by computer. Our search builds a profile one step at a time, always making next the choice with the fewest options left open to it, and keeping for every choice still outstanding a running list of the options that remain consistent with what has been fixed. Together with the symmetries of the tournament itself this bring within reach of a single laptop instances that neither integer programming nor a general-purpose SAT solver can decide. The refutations at orders 19 and 23 are also formally certified: the search is split into independent subproblems, each is refuted by a SAT solver that emits a machine-checkable proof, and a separate program rechecks every one of them. The whole Paley(23) refutation took **22 hours on one laptop**, on the instance quoted above as undecided after six weeks. Everything is reproducible from a public repository, with only two simple combinatorial lemmas remaining human rather than machine-checked.

1. Introduction

Let T be a tournament on the vertex set V , $|V| = n$: a digraph in which every unordered pair $\{i, j\}$ carries exactly one of the arcs $i \rightarrow j$ or $j \rightarrow i$. A k -profile consists of k linear orders $\pi_1, \dots, \pi_k$ on V . The *majority tournament* of such a profile puts an arc from i to j whenever $|\{v : i \prec_{\pi_v} j\}| > k/2$. For odd k this is well defined on every pair, and the result is a tournament. We say T is **k -inducible** if some k -profile has T as its majority tournament, and we call such a profile a *witness*.

Call the number of voters who rank i above j the **support** of the arc $i \rightarrow j$. In a witness every arc's support exceeds $k/2$, so for $k = 5$ it is 3, 4 or 5, and it is useful to grade inducibility by how decisively the arcs are carried: a witness has **margin at most** M if every arc has support s with $2s - k \leq M$. For $k = 5$ the possible margins are 1 (every arc carried exactly three votes to two, which we call *unit margin*), 3 (every support in $\{3, 4\}$) and 5 (no restriction, i.e. plain majority). Unit margin is a strict restriction, so a unit-margin witness is in particular a majority witness, while a unit-margin *refutation* is strictly less informative than a majority one.

McGarvey's theorem [16] guarantees that every tournament is k -inducible for k large enough, and the least such k is the *majority dimension* of T , written $\dim(T)$. The inverse question - which tournaments are reachable with a *fixed* small number of voters — is the subject of this paper.

Counting arguments show that almost all tournaments fail to be k -inducible for any fixed k , yet the *explicit* examples they yield are astronomically large, and it is small explicit examples that structural results need. This paper closes much of that gap for $k = 5$.

Let $N(k)$ be the least order of a tournament that is not k -inducible. The answer is only known exactly for $k = 3$. For $k = 5$ the state of the art before this work was

$$12 \leq N(5) \leq 38,$$

both ends of which are due to our companion paper [2]: the lower bound from its exhaustive census of all $D_{11} \approx 9.0 \times 10^8$ tournaments on 11 vertices, of which about 3.8×10^8 are even 3-inducible and every one of the remaining 5.2×10^8 admits a margin-1 five-voter profile. Bachmeier et al. [1] had previously reached $n \leq 10$ exhaustively, giving $N(5) \geq 11$.

We spell out the upper end because the bounds in the literature are of two different kinds and they are not comparable. Some are obtained by counting: one compares the number of tournaments of a given order with the number that k linear orders can produce, and when the first exceeds the second, some tournament of that order must be missed. Such an argument names no example. Bachmeier et al. [1] argued this way to reach 41, and [2] sharpened the comparison by restricting it first to regular and then to near-regular tournaments, reaching 39 and then 38. The other kind of bound exhibits a tournament and checks it. There the record was 603 979 799 vertices [1], from a dominating-set theorem of Alon et al. [4] together with a construction of Graham and Spencer [5], and then 43 [2] — Paley(43), the first example of moderate order.

Our $N(5) \leq 23$ exhibits a tournament, and it is below both lines at once: it improves the best exhibited bound from 43 to 23, and since $23 < 38$ it also beats the counting-based upper bound. It is obtained by a bespoke algorithmic decision procedure that fits this problem particularly well.

1.1 Why Paley(23) is hard to classify

The Paley tournament P_q , for a prime $q \equiv 3 \pmod{4}$, has vertex set $\mathbb{F}_q$ and an arc from i to j whenever $j - i$ is a nonzero square. It is *doubly regular*: every vertex has out-degree $(q - 1)/2$ and every ordered pair of vertices has exactly $(q - 3)/4$ common out-neighbours. Its automorphism group — the relabellings of the vertices carrying every arc to an arc, written $\text{Aut}(P_q)$ and defined with the rest of our group-theoretic notation in Section 2 — has order $q(q - 1)/2$, which equals the number of arcs, so exactly one automorphism carries any given arc to any other.

Paley tournaments are the natural candidates for non-inducibility because they are far from being transitive, and because their symmetry makes an otherwise hopeless search merely very hard. Bachmeier et al. [1] call them *quadratic residue tournaments* and write Q_p ; their orientation is the converse of ours, which costs nothing, because reversing every voter's order reverses every arc of the majority tournament and changes no support. A tournament and its converse therefore have the same majority dimension and are inducible at exactly the same margins. We use the name Paley and the notation P_q throughout.

Their SAT study establishes

$$\dim(Q_3) = \dim(Q_7) = 3, \quad \dim(Q_{11}) = \dim(Q_{19}) = 5,$$

where $\dim$ denotes majority dimension, and then stops exactly where the problem becomes hard:

“Unfortunately, we were not able to check whether the majority dimension of Q_{23} is equal to 5 or larger as the SAT solver did not terminate within a total of six weeks.”

That sentence is the point of departure for this paper.

1.2 Why Paley(43) was easy to classify

By contrast, our earlier proof that Paley(43) is not 5-inducible [2] never searched the profile space in general. It rests on a quantity we call the *slack*,

$$\text{slack}_5(T) = 5 \text{MAS}(T) - 3C, \quad C = \binom{n}{2},$$

where MAS is the maximum number of arcs of T that a single linear order can agree with. Summing the majority condition over all arcs bounds how far the five voters can *collectively* fall short of that maximum, by exactly $\text{slack}_5(T)$. A small slack therefore pins the voters into a shallow shell around the best orders, and at $q = 43$, where $\text{slack}_5 = 6$, that shell is small enough to enumerate outright — about 1.8 million cases once its symmetries are quotiented out — which is how [2] settles the case without searching the profile space at all.

The essential point is that **this is not a threshold in q** . The argument is valid at every q ; what varies is whether the shell it must enumerate is small enough to enumerate. Slack controls the depth of that shell, and it is *not* monotone in q :

q	7	11	19	23	27	31	43
slack_5	7	10	22	46	27	30	6

Paley(43) has the smallest slack anywhere in the family, and Paley(23) has nearly eight times as much. At slack 46 the forced level is far deeper than 1 and the shell explodes combinatorially, so the screen is not so much silent at $q = 23$ as unusable there. As [2] puts it, the proof “exploits a genuine numerical coincidence”. Nothing in that approach descends toward $N(5)$, and no amount of computation would make it.

One quantity, two obstructions. It is worth separating two arguments that this table runs together, because they concern different methods even though they are governed by the same number. Writing $\text{slack}_5(T) = 5 \text{MAS}(T) - 3C$, and noting that $\alpha^* = \text{MAS}/C$ throughout this family, the two are related by

$$\text{slack}_5 = 5C \left(\alpha^* - \frac{3}{5} \right).$$

Read as *slack*, a large value says the screen of [2] must enumerate a deep shell, and the enumeration is what becomes impossible. Read as *predictability*, the same large value says the necessary condition $\alpha^* \geq 3/5$ is far from binding, so **no argument resting on that linear relaxation can certify the instance** — not the screen, and not a linear program either. The second reading is the more damaging of the two, and it is what makes a decision procedure necessary rather than merely convenient.

It is also what separates $k = 5$ from $k = 3$. At $k = 3$ predictability decides exactly at $N(3)$: the 96 obstructions on eight vertices have $\alpha^* = \frac{13}{20}$, which is *below* $\frac{2}{3}$, so the relaxation detects them and $N(3) = 8$ follows. At $k = 5$ it does not: the table above has $\alpha^* > \frac{3}{5}$ at every q , yet Paley(23), Paley(27), Paley(31) and Paley(43) are all non-inducible — Paley(23), which supplies our bound, sits $\frac{2}{55}$ *above* the threshold. The failure of sufficiency therefore arrives before $N(5)$, whereas at $k = 3$ it arrives after $N(3)$: there the first instance where $\alpha^* \geq \frac{2}{3}$ is not sufficient has eleven vertices [2], comfortably past $N(3) = 8$, and its threshold is attained only by a distribution whose denominator does not divide three. Predictability will eventually fall below $\frac{3}{5}$, since $\alpha^* \leq \text{MAS}/C \rightarrow \frac{1}{2}$ for almost all tournaments, and any instance where it does is immediately non-inducible. But that crossing happens strictly later than $N(5)$, so it cannot be the route to it.

The gap between these two situations is the subject of this paper: the interesting instances are precisely those too large for exhaustive profile search, too symmetric for SAT, and too slack for a counting screen.

1.3 Contributions

1. $N(5) \leq 23$ (Section 3.1). Paley(23) is not 5-inducible. This settles the instance on which the SAT solver of [1] ran for six weeks without terminating. It improves the best exhibited bound from 43 to 23, and since $23 < 38$ it is also below every bound reached by counting, so nothing in the bracket now rests on an argument that names no example. We also decide Paley(27) and Paley(31), which do not improve the bound but demonstrate the method’s reach: each of the two larger tournaments is settled in less time than Paley(23).
2. $N(5) \geq 13$ (Appendix C). Every tournament on 12 vertices is 5-inducible. The analysis settles all 2,048 one-vertex extensions of each of the D_{11} tournaments on 11 vertices, which is complete because every 12-vertex tournament is such an extension — of each of its twelve one-vertex deletions.
3. **A formally certified negative result** (Appendix B.1). Paley(19) is not margin-1 5-inducible. The search is split into independent subproblems (*cubes*); each is refuted by the SAT solver CaDiCaL, which emits its refutation as a machine-checkable *LRAT* proof; and every one of those proofs is then rechecked by a separate program that shares no code with the solver. We publish two root hashes and state precisely which one a third party must reproduce.
4. **Paley(43) is not vertex-critical** (Section 3.3). Deleting a vertex from Paley(43) leaves a 42-vertex tournament that is still not 5-inducible. Since Paley(43) is vertex-transitive, one computation settles all 43 deletions.

1.4 Related work

Two papers stand immediately behind this one, in different ways. Our companion paper [2] is the prior art for the *bound*: it produced the first explicit non-5-inducible tournament of modest size, Paley(43), and its two bounds are what we measure against: $N(5) \leq 38$ by counting and $N(5) \leq 43$ explicitly. Bachmeier et al. [1] is the prior art for the *problem and the method*: it introduces the computational study of k -majority digraphs, characterises the 3-majority case building on Dushnik and Miller [9], and establishes hardness results for voting with a constant number of voters. Their explicit non-5-inducible tournament rests on the dominating-set function F of Alon et al. [4], on Fidler’s bound of 12 for five voters [6], and on Graham and Spencer’s construction [5]. Both [4] and [6] index F by the majority parameter rather than the voter count, so the bound they state is $F(3) \leq 12$; we quote it by voter count throughout, following [1]. The case $k = 3$ is settled, and it is worth separating the two halves. Shepardson and Tovey [8] introduced the *predictability* $\alpha^*(T)$: the largest threshold α for which T can be realised by a profile in which every arc is supported by at least an α fraction of the voters. A 3-voter profile is exactly a $\frac{2}{3}$ -supermajority representation, so $\alpha^*(T) < \frac{2}{3}$ rules out 3-inducibility; their 8-vertex tournament has $\alpha^* = \frac{13}{20}$, which gives $N(3) \leq 8$ at once. Our earlier paper [2] refuted their conjecture that $\alpha^* \geq (m+1)/2m$ suffices for m -inducibility. The lower bound needs more, because the converse implication is false — [2] exhibits an 11-vertex tournament with $\alpha^* = \frac{2}{3}$ that is not 3-inducible — so $\alpha^* \geq \frac{2}{3}$ throughout $n \leq 7$ does not settle it. Eggermont, Hurkens and Woeginger [7] close that half by direct computation: every tournament on 7 vertices is 3-inducible, exactly 96 of the 6,880 on 8 vertices are not, and every tournament on 8 or 9 vertices is 5-inducible. Hence $N(3) = 8$ and $N(5) \geq 10$. The count of 96 is independently reported by [1], and our engine reproduces it as a known-answer test. Milosz, Hamel and Pierrot [17] proved that for three voters, every minimum-weight feedback arc set of the majority tournament is a minimal hitting set of its directed 3-cycles; [2] strengthens this to arbitrary tournaments and refutes both of their conjectures.

Our certified computation (Appendix B.1) is standard in shape: it splits the problem into independent subproblems, in the manner of *cube-and-conquer* [10] — a SAT instance is split by fixing a few variables at a time, giving many partial assignments called *cubes*, and each cube is handed to the solver as a problem of its own — and it adopts the standard of evidence set by the Boolean Pythagorean triples proof [3, 12], in which every subproblem emits a machine-checkable refutation that an independent checker validates. Our refutations are in the LRAT format [11], where [3] emitted DRAT and [12] used its verified GRIT pipeline. Appendix B.1 says what we take from that line of work and in which two respects our artifact is weaker.

2. Preliminaries

Throughout, T is a tournament on $V = \{1, \dots, n\}$ and $k = 5$ unless stated otherwise. We write $N^+(v)$ and $N^-(v)$ for the out- and in-neighbourhoods of v . Vertices are numbered from 1 in all prose and tables; the software numbers them from 0, and the shift is applied consistently in the repository's verification scripts.

The *support* of an arc $e = (i \rightarrow j)$ under a profile is the number of voters ranking i above j , written $c(e)$. A profile is a witness for T when $c(e) \geq (k+1)/2$ for every arc, and the *margin* of e is $2c(e) - k$. The three regimes for $k = 5$ are margin 1 (every $c(e) = 3$), margin 3 ($c(e) \in \{3, 4\}$) and margin 5 (no constraint beyond being a witness).

Several families swept here are large enough that their sizes are better named than written out. Following [2], D_n , R_n and S_n denote the numbers of isomorphism classes of, respectively, all tournaments, regular tournaments and self-converse tournaments on n vertices. The text gives these to two significant figures; Appendix D tabulates the exact values, which are what the completeness gates of Appendix C are checked against.

2.1 Base states

Every search here starts by fixing how the voters rank a handful of vertices and then looks for ways to fit the rest around that. Choose a *base* $B \subseteq V$ of a few vertices — five or six in practice — and write $T|_B$ for the subtournament induced on B . A **base state** is one way for the k voters to rank B that is already consistent with what T demands there: k orderings of B whose majorities reproduce $T|_B$ at the margin being asked for. Base states differing only by a relabelling of the voters are counted once, which Lemma 2.3 shows is legitimate.

Base states are the unit of work, and the loop over them is the **outer loop**. Each is a self-contained subproblem — the rankings of B are given, and the question is whether the remaining $n - |B|$ vertices can be inserted so that every arc of T comes out right — so the search splits across base states with no communication, and T fails to be k -inducible exactly when every base state fails. How many there are depends only on $|B|$, on $T|_B$ and on the margin regime, and *not* on n : with the five-vertex base we use on the Paley tournaments there are 8,031 of them whether the tournament has 23 vertices or 47. Enlarging the base therefore adds work inside each subproblem without creating more of them.

2.2 Symmetry reductions

An *automorphism* of a digraph D is a permutation of its vertices carrying every arc to an arc; these form $\text{Aut}(D)$ under composition, and D is **rigid** when $\text{Aut}(D)$ is trivial. For a permutation group Γ on a set X , the **orbit** of x is $\Gamma x = \{\gamma(x) : \gamma \in \Gamma\}$, and the orbits partition X . We use orbits of $\text{Aut}(T)$ on the vertices, on the arcs, and on the **ordered pairs** of distinct vertices, where γ sends (u, v) to $(\gamma(u), \gamma(v))$. An ordered pair is either an arc or the reverse of one, so that third action has at least two orbits.

Two symmetries act on witnesses and both are exploited by every method here: the $k!$ relabellings of the voters, and $\text{Aut}(T)$ acting on the vertices. Each yields a reduction without loss of generality. The point of the second is that a witness can be pushed around by an automorphism, so we may decide in advance what some voter's top two vertices are.

Lemma 2.1 (orbit anchoring). *Let $\text{Aut}(T)$ act on the ordered pairs of distinct vertices, and choose one representative pair from each orbit, say $(a_1, b_1), \dots, (a_m, b_m)$. If T is k -inducible, then T has a witness in which, for at least one index i , some voter ranks a_i first and b_i second.*

Proof. Let $\pi_1, \dots, \pi_k$ be a witness and let (u, w) be the top two vertices of π_1 . The pair (u, w) lies in one of the orbits; let (a_i, b_i) be that orbit's representative, so there is $g \in \text{Aut}(T)$ with $g(u) = a_i$ and $g(w) = b_i$.

Applying g to every voter gives a profile whose majority tournament is $g(T) = T$, so it is again a witness, and its first voter ranks a_i first and b_i second. ■

So we may search only for witnesses whose top pair is one of the m chosen pairs, a saving of $n(n-1)/m$. The price is that **all m representatives must be searched**: the lemma guarantees that *some* i works, not any particular one, so a run covering only one representative proves nothing. For a Paley tournament $m = 2$, since an ordered pair is either an arc or a non-arc and $\text{Aut}(P_q)$ is transitive on each set. The two representatives can be exhibited without computing the group: $x \mapsto a(x-t)$ with a a quadratic residue sends (t, s) to $(0, a(s-t))$, and $a(s-t)$ sweeps the coset $\text{QR} \cdot (s-t)$, so one representative per coset suffices, the two cosets being mutually exclusive and exhaustive.

Corollary 2.2 (vertex anchoring). *Let $\text{Aut}(T)$ act on the vertices and choose one representative $v_1, \dots, v_r$ from each orbit. If T is k -inducible, then T has a witness in which some voter ranks some v_i first.*

Proof. The proof of Lemma 2.1 verbatim, with the action on vertices in place of the action on ordered pairs. ■

A witness meeting either restriction is called **anchored**. The two are alternatives rather than a sequence, each without loss of generality, and which is better depends on the tournament: for a vertex-transitive one the vertex form says only that some voter ranks vertex 1 first, a factor of about n , against $|\text{Aut}(T)|$ for the pair form — but the pair form costs more to test at every node, and where the vertex stabilisers are trivial the vertex form is already as strong as anchoring can be.

Lemma 2.1 is only as good as the orbits it is given. It licenses the restricted search *provided* the representatives meet every orbit. Suppose they miss one. Then every witness anchored in the missed orbit is discarded unexamined, and the search can report that no witness exists when one does — the one error that would matter here, since the results we are after are negative. Nothing inside the search can detect this, because from the inside a discarded branch and an impossible branch look alike. The check therefore happens outside: for every tournament we compute $\text{Aut}(T)$ separately, with **nauty** [20], and record its orbits alongside the run. Using the orbits of a known subgroup $\Gamma \leq \text{Aut}(T)$ is safe rather than risky, since Γ -orbits are finer and a representative set meeting all of them meets every $\text{Aut}(T)$ -orbit; the only cost is searching more than we had to.

Lemma 2.3 (voter lex-ordering). *If T is k -inducible, then T has a witness whose k orders are non-decreasing under any fixed total order on linear orders.*

Proof. The support $c(e)$ counts voters and is therefore invariant under permuting them, so any profile with the same multiset of orders induces the same tournament. Sorting the multiset gives the required witness. ■

Together the two lemmas reduce the running time by more than two orders of magnitude, and they bite in different places. Lemma 2.3 acts first and is absorbed into the definition of a base state: the k orderings of B are held sorted and each multiset counted once, dividing the count by a factor approaching $k! = 120$. Lemma 2.1 acts second and *deletes* base states outright, since a base state can extend to an anchored witness only if one of its voters already ranks a chosen representative pair at the top of B — a property of the base state alone, testable before any search begins. What survives is **live**. For the five-vertex base on the Paley tournaments the representatives we run leave 2,591 of the 8,031 live, and the best available choice would leave 2,537; the remaining three quarters are discarded for free.

What makes the two lemmas compose rather than conflict is that every condition above is phrased as “*some* voter ranks ...”, never “voter 1 ranks ...”. A condition on a particular voter would be destroyed by the re-sorting of Lemma 2.3, whereas one invariant under permuting the voters survives it, so the reductions multiply. Both are available to every formulation compared in Appendix A, which is what makes that a comparison of search strategies rather than of symmetry handling; the routes differ only in *when* the Lemma 2.1 condition is tested. Our depth-first search tests it inside the search rather than at the leaves, which is sound because the condition is monotone: once a partial profile has lost the chance to satisfy it, inserting further vertices cannot restore it. The SAT route tests it once, at the root, where it decides which cubes reach the solver at all.

2.3 The search in five steps

The engine extends base states. A base state fixes how the k voters rank B ; the search inserts the remaining $n - |B|$ vertices into those rankings one at a time, backtracking as soon as the arcs already decided contradict T . A base state extended to k full rankings yields a witness; one that cannot be is eliminated; and T fails to be k -inducible exactly when every base state is eliminated. Read as a pipeline, with T and the margin regime as inputs:

1. **Pick a size.** Choose $b = |B|$, in practice 5 or 6: the work of steps 3 and 4 grows steeply in b while the depth it removes from step 5 grows only linearly.
2. **Pick a subtournament.** Choose B with $|B| = b$. Only $T|_B$ matters for what follows, so the choice can be optimised cheaply (Appendix A.4) — and should be, since the base-state count varies by more than two orders of magnitude across the twelve isomorphism classes at $b = 5$.
3. **Find its base states.** Enumerate the k -tuples of linear orders of B whose supports agree with $T|_B$, counted up to voter permutation by Lemma 2.3.
4. **Determine the usable symmetries.** Compute $\text{Aut}(T)$, or a subgroup, and its orbits, then impose the ordered-pair anchoring of Lemma 2.1, the vertex anchoring of Corollary 2.2, or neither, discarding the base states no anchored witness can use. Imposing neither is sometimes right: at unit margin the anchoring rules out so few partial profiles that it does not repay testing at every node, and anchoring is the only step that discards profiles at all — a risk not worth running when the verdict being chased is negative.
5. **Run to completion, in parallel across base states.** The verdict is positive as soon as any base state yields a witness. It is negative only when *every* live base state has been searched to exhaustion and no search was stopped by its node or time cap. A run that hit a cap is not a verdict.

Steps 1–4 are cheap and are what a human chooses; step 5 is the computation. The same five steps describe the SAT route, with step 5 replaced by handing each live base state to a CDCL solver as a *cube* — the partial assignment fixing the voters’ rankings of B — so the two share their decomposition and their two lemmas and differ only in what refutes a leaf.

2.4 Paley tournaments of prime-power order

For $q \equiv 3 \pmod{4}$ a prime power, P_q is as defined in Section 1.1. For prime q the construction over $\mathbb{Z}/q$ is valid; for a prime power such as $q = 27$ it is not, and the tournament must be built over $\mathbb{F}_q$. We note this because the naive $\mathbb{Z}/27$ construction silently fails to be a tournament — 81 pairs receive no arc — and searching such an object produces confident nonsense; our implementation refuses it. Built over $\mathbb{F}_{33}$, the tournament is, by canonical form, the unique most symmetric doubly regular tournament on 27 vertices.

The automorphism count also changes, and Section 1.1 states it for primes only. In general $|\text{Aut}(P_q)| = q(q-1)e/2$ with $e = [\mathbb{F}_q : \mathbb{F}_p]$, since the field automorphisms contribute, so the coincidence with the arc count $q(q-1)/2$ is special to prime q . At $q = 27$ the group has order 1053 and is still transitive on arcs and on non-arcs, which is all Lemma 2.1 needs, and that transitivity holds at every q ; we verified all of these group orders independently.

3. Results

Two regimes recur below, so we recall them. *Unrestricted* (or plain majority) inducibility asks only that each arc be carried by three, four or five of the five voters. *Unit margin* asks that every arc be carried by exactly three against two — a strict restriction, so a unit-margin witness is also an unrestricted one, while a unit-margin refutation is weaker than an unrestricted one. Only an unrestricted refutation bounds $N(5)$.

All runs in this section were performed on a single laptop (Apple M-series, 14 cores, 24 GB RAM), using at most 11 cores concurrently, with three exceptions that used a compute cluster: the census of regular tournaments on 15 vertices below, the self-converse census of order 13 below, which was split between the two machines, and the computation of Appendix C. Costs are collected in one table at the end of this section and reported in *core-hours*, the product of wall time and the number of workers; they are not repeated in the text. Core-hours are comparable across our own runs but are **not** machine-independent: Appendix A.5 records this laptop running 2.6 to 3.3 times faster per core than the cluster we also used on the computations measured there, and 5.1 times faster on the self-converse census, so the ratio is workload-dependent and not a constant to extrapolate with. The quantity that does not depend on the machine is the *node count*, which we give wherever it was recorded; Section 5.3 sets out exactly what a replication must hold fixed for it to match.

3.1 Paley(23), Paley(27) and Paley(31) are not 5-inducible

Theorem. *Paley(23) is not 5-inducible. Consequently $N(5) \leq 23$.*

The proof is a search that runs to completion. The base is $\{1, 2, 3, 6, 12\}$, which gives 8,031 base states; each is searched exhaustively for a five-voter profile inducing Paley(23); and none contains one. The pieces were run and logged separately, which lets us verify three things afterwards that a single aggregate figure would hide: that the pieces between them account for every base state exactly once, neither missing one nor counting one twice; that no piece was stopped early by a limit on time or on the size of its search; and that no piece reported a profile. The file of witnesses is empty.

The same method settles two larger cases, checked the same way: Paley(27) is not 5-inducible, over the same 8,031 base states, and Paley(31) is not 5-inducible, over 21,009 base states with a different base. In both, the logs account for every base state exactly once, nothing was stopped early, and no profile was found. Appendix A.5 gives the costs.

Neither Paley(27) nor Paley(31) improves the bound on $N(5)$. Since 5-inducibility is hereditary, any tournament containing a non-inducible subtournament is itself non-inducible, so non-inducible tournaments already exist at every order ≥ 23 ; exhibiting two more at 27 and 31 adds nothing to the bound. Their value is *method reach* — they show that the approach decides 27- and 31-vertex instances on one laptop, where SAT and integer programming do not reach at all.

Criticality across the family

Since arc-criticality implies vertex-criticality (Section 3.3), each cell below records only the *strongest* property established, at each of the two margins. “Not vertex-critical” therefore also rules out arc-criticality.

q	margin ≤ 1	unrestricted
19	arc-critical	5-inducible, so criticality does not apply
23	not vertex-critical	arc-critical (Section 3.4)
27	not vertex-critical	open
31	not vertex-critical	open
43	not vertex-critical	not vertex-critical (Section 3.3)

Three remarks. Paley(19) is the only member critical at margin 1, and it is critical in the strongest sense: reversing any one arc makes it margin-1 inducible, which by the implication also gives vertex-criticality. At the other end, Paley(43) fails to be vertex-critical at *both* margins, so the obstruction there is not concentrated at a vertex in either regime.

The interesting cell is $q = 23$: it is **vertex-critical unrestricted** — deleting any vertex leaves a 5-inducible tournament, and Paley(23) is vertex-transitive so one deletion settles all 23 — while being **not** vertex-critical at margin 1, since Paley(23) $- v$ is not margin-1 inducible either. Criticality is therefore not monotone in the margin, and a tournament can be a minimal obstruction in one regime and not in the other.

The two open cells are open by choice, not by obstacle, and two separate questions sit behind each. Vertex-criticality asks whether the tournament minus a vertex is 5-inducible; both Paley(27) and Paley(31) are vertex-transitive, so a single deletion decides all q of them, and the computation has the same shape as the Paley(43) $- v$ sweep of Section 3.3. Arc-criticality is the more expensive question: reversing an arc destroys the symmetry that makes these tournaments tractable, so every base state becomes live and the scan is a large multiple of the $q = 23$ one (Section 3.4). Neither answer would move $N(5)$ or bear on any other statement here, so we have not run them.

3.2 Paley(19) is 5-inducible, but not with unit margin

That Paley(19) is 5-inducible is due to Bachmeier et al. [1], who computed $\dim(Q_{19}) = 5$. Our search recovers it, and we verify the witness independently with a script that shares no code with the search: it reads the witness and the tournament and confirms that all 171 arcs are supported by at least 3 of the 5 orders. We record the agreement because it is the cheapest available check that our engine and their encoding are deciding the same problem.

Our contribution at $q = 19$ is the negative half, and its certificate.

It is *not* 5-inducible with unit margin. This is a complete refutation over all 2,200 base states of the base $\{1, 2, 3, 4, 6\}$, and it is the statement we certify formally in Appendix B.1.

The two facts together locate Paley(19) precisely on the margin scale: it is inducible, but every witness must carry some arc by 4–1 rather than 3–2. Indeed the witnesses are not delicate. A shuffled ten-way sweep at margin ≤ 3 found witnesses in all ten workers simultaneously, from ten different base states, with no base state refuted; the support histogram of one such witness is 159 arcs at 3–2 and 12 at 4–1, with none unanimous.

Both negative results in this section are additionally certified formally, by cube-and-conquer computations in which every leaf is refuted by CaDiCaL [18, 24] in LRAT and rechecked by `lrat-trim` [24], a separate program that shares no code with the solver, with the solver’s own proof checking deliberately disabled. Both are complete on *both* halves — the leaves and the claim that the leaves are exhaustive — and both reduce to the same two human lemmas, 2.1 and 2.2. Appendix B gives the construction, the published root hashes, the two respects in which our artifact is weaker than the Boolean Pythagorean triples proof it is modelled on, and one episode in which the certificate asserted more than it had established.

3.3 Paley(43) is not vertex-critical

Call a tournament T *vertex-critical* for k -inducibility if T is not k -inducible but $T - v$ is k -inducible for every vertex v . Vertex-criticality is the natural minimality notion for an obstruction, and Paley(43) — the first explicit non-5-inducible tournament [2] — is the natural candidate.

Theorem. *Paley(43) minus a vertex is not 5-inducible. Hence Paley(43) is not vertex-critical.*

Since Paley(43) is vertex-transitive, all 43 one-vertex deletions are isomorphic and a single computation settles them all. The instance has $n = 42$; the sweep is complete over all 8,031 base states of the base $\{1, 2, 3, 4, 11\}$, using the orbit break of Appendix A.3 with the two representatives described below. The audit is exact: markers for base-state indices $0, \dots, 8030$ with none missing, none duplicated, nothing capped or aborted.

The orbit break. $\text{Aut}(P_{43} - v)$ contains the stabiliser of v in $\text{Aut}(P_{43})$, which is cyclic of order 21 and splits the remaining 42 vertices into exactly two orbits of size 21 — the quadratic residues and the non-residues. Two representatives, one from each, therefore meet every orbit; and since the stabiliser is a subgroup of the full automorphism group of the deleted tournament, its orbits refine the true ones and the check is conservative.

Why a margin-3 sweep settles the unrestricted question. The computation was run at margin ≤ 3 , which is cheaper than the unrestricted problem. For this tournament the two coincide, by the following observation.

Lemma (3-cycle bound). *Let $u \rightarrow v \rightarrow w \rightarrow u$ be a directed triangle of T and let $\pi_1, \dots, \pi_k$ be any linear orders. Then*

$$c(u \rightarrow v) + c(v \rightarrow w) + c(w \rightarrow u) \leq 2k.$$

Consequently, in any majority witness, every arc lying in a directed triangle has support at most $2k - 2\lceil(k+1)/2\rceil$, which is 4 when $k = 5$.

Proof. A linear order cannot rank u above v , v above w and w above u simultaneously, so each voter agrees with at most two of the three arcs; summing over the k voters gives the inequality. If $c(u \rightarrow v) = 5$ then the other two supports sum to at most 5, contradicting the majority requirement that each be at least 3. $\square$

Every arc of $P_{43} - v$ lies in at least 10 directed triangles — indeed no arc lies in none — so no arc of a majority witness could be unanimous, and margin ≤ 3 is not a restriction at all. The margin-3 refutation is therefore an unrestricted refutation.

The same observation applies to every tournament in this paper, and more generally: in any *regular* tournament every arc lies in a triangle, because for an arc $u \rightarrow v$ both $N^+(v)$ and $N^-(u)$ are subsets of $V \setminus \{u, v\}$ of size $(n-1)/2$ each, so they intersect. We use the lemma only where stated.

A corollary: Paley(43) reconfirmed independently. Since 5-inducibility is hereditary on subtournaments, a non-inducible subtournament forces the whole tournament to be non-inducible. Paley(43) $- v$ is a subtournament of Paley(43), so the theorem above re-derives the main result of [2], $N(5) \leq 43$, *by a different method*: a complete decision procedure over the profile space rather than the counting screen of [2], sharing no code and no argument with it. We record this because a computational bound that has been obtained only once, by only one method, is worth less than the same bound obtained twice; the reader who doubts the co-backing screen of [2] can take Paley(43) from here instead.

What the theorem does not give. It does not improve $N(5) \leq 23$: the instance has 42 vertices. Its content is that the obstruction in Paley(43) is not concentrated at any single vertex, so the search for a minimal obstruction must descend further.

3.4 Vertex and arc criticality for Paley(23)

Theorem. *Paley(23) is arc-critical: reversing any single arc yields a 5-inducible tournament. Consequently it is vertex-critical.*

Let T^e denote Paley(23) with the arc $(1, 2)$ reversed. One instance settles all 253 arcs, and the reduction is an explicit orbit calculation rather than an appeal to symmetry in general: the group $\{x \mapsto ax + b : a \text{ a nonzero square}\}$ has order $253 = q(q-1)/2$, every one of its elements preserves the arc set, the orbit of $(1, 2)$ is the full arc set, and the stabiliser is trivial. So $\text{Aut}(P_{23})$ is transitive on arcs, and any two single-arc reversals of Paley(23) are isomorphic.

The computation is a scan of the 8,031 base states of T^e for a 5-voter profile inducing it. Reversing an arc destroys all symmetry, $|\text{Aut}(T^e)| = 1$, so no break is available and every base state is live. A witness was found at base state 1,161, after which the scan was stopped; the remaining 6,996 states were never examined, which is sound here precisely because a witness is a positive certificate — completeness of the enumeration is what a negative verdict needs, and this is not one.

The witness was verified independently of the search that produced it, by three checks on the five ballots alone: that each is a permutation of the 23 vertices; that the majority tournament they induce agrees with T^e in every one of the 253 arcs and therefore differs from Paley(23) in exactly the reversed one; and that the support of every arc lies in $\{3, 4\}$. The last is a consistency check against the 3-cycle bound of Section 3.3 rather than a requirement: no arc is unanimous, as none can be, since every arc of T^e lies in at least five cyclic triangles.

Two consequences. First, since arc-criticality implies vertex-criticality, this reproves by a different route the vertex-criticality already established directly. Second, it does *not* extend to margin 1: the witness has arcs at support 4 as well as 3, and independently, Paley(23) is not vertex-critical at margin 1, so by the same

implication it cannot be arc-critical there either. Paley(23) is thus arc-critical unrestricted and not even vertex-critical at unit margin — the sharpest instance in the family of the non-monotonicity noted in Section 3.1.

Vertex-critical but maximally not arc-critical. Arc-criticality implies vertex-criticality, so it is natural to ask whether the two coincide. They do not, and the gap is as wide as it can be. Let h be one of the four unit-margin obstructions among the 110 vertex-transitive tournaments on 21 vertices (Section 3.5). Then h is *vertex-critical* at unit margin: $h - v$ has a unit-margin witness, and since $\text{Aut}(h)$ is transitive on the vertices, all 21 deletions are isomorphic and one computation settles every one of them. Yet no arc reversal helps at all. Trivial vertex stabilisers force trivial arc stabilisers, so the 210 arcs fall into exactly ten orbits of size 21, and each of the ten representatives is refuted by an exhaustive sweep of its 2,200 base states — 736 separate runs in total, every one exhausted with no cap and no witness, tiling each tournament’s range exactly. So all 210 single-arc reversals of h remain unit-margin obstructions. Deleting a vertex always restores inducibility; reversing an arc never does.

Note the contrast with Paley(23), which is arc-critical unrestricted: there *every* arc reversal restores inducibility. The two notions therefore come apart in both directions across the tournaments we have, and neither implies anything about the other beyond the one implication above.

And arc-criticality is not a property of the tournament. All four unit-margin obstructions among the 110 have $|\text{Aut}| = 21$ with trivial stabilisers, so each has exactly ten arc orbits of size 21; we take as representative the lexicographically least arc of each orbit. Sweeping those representatives at unit margin gives four different pictures, and they line up with the group rather than with the arcs:

tournament	Aut	the ten orbits	obstructions
h_1	$\mathbb{Z}_{21}$	U U U U U U U U U U	10
h_2	$\mathbb{Z}_{21}$	U S U S S U U S S U	5
h_4	$\mathbb{Z}_7 \rtimes \mathbb{Z}_3$	S S S S S S S S S S	0
h_5	$\mathbb{Z}_7 \rtimes \mathbb{Z}_3$	S S S S S S S S S S	0

U marks a reversal that is still not unit-margin inducible and S one that is. The rows are to be read as multisets: the number of orbits, their common size and the verdicts they carry are isomorphism invariants, but the order in which the ten appear is not, since the lexicographic rule is relative to each tournament’s supplied vertex numbering and only two of the four are labelled canonically. Nothing below compares orbit i of one tournament with orbit i of another.

Three things follow. First, **arc-criticality at unit margin is carried by the arc and not by the tournament**: h_2 has five orbits whose reversal stays an obstruction and five whose reversal does not, interleaved, so no statement of the form “this tournament is or is not arc-critical” captures it. Since every orbit has size 21, that is exactly 105 of its 210 arcs whose reversal restores inducibility and 105 whose reversal does not; we call such a tournament **arc-semi-critical**. The three possibilities are then all realised among the tournaments we have: Paley(23) is arc-critical at unrestricted margin, every one of its arcs being critical; h_2 is arc-semi-critical; and h_1 has no critical arc at all. Second, h_1 and h_2 sit at the two extremes of that scale, 10 of 10 and exactly 5 of 10, with nothing between them among the tournaments we have. Third, and the reason we report the automorphism groups: **every one of the 15 reversals that remains an obstruction belongs to a tournament with cyclic automorphism group**, and all 20 orbits of the two with the nonabelian group of order 21 are inducible. That is exhaustive over these four tournaments — 40 orbits, no undecided cases — but it is four tournaments, two of each kind, and we have no mechanism to offer for it; we record the alignment rather than claim a law.

The picture at 21 vertices, in summary. Taken together with the vertex deletions, the four tournaments settle the question of how far the two notions of criticality can come apart inside a single family, and the answer is: as far as possible in one direction and not at all in the other. Every unit-margin obstruction we know at 21 vertices is *vertex-critical*, and exhaustively so: the four tournaments and the 15 arc reversals that remain obstructions have 319 one-vertex deletions between them, 289 of them pairwise non-isomorphic

by canonical form, and every one of the 289 has an exhibited unit-margin witness. There is no exception anywhere. Arc-criticality, by contrast, takes all three of its possible values among four tournaments of the same order with automorphism groups of the same size. Deleting a vertex therefore always restores inducibility at these orders, while reversing an arc restores it always, half the time, or never, according to the tournament. A corollary worth stating for its own sake: since no one-vertex deletion of any of them is an obstruction, none of these tournaments descends to a unit-margin obstruction on 20 vertices. Neither does anything at 19: both doubly regular tournaments there are vertex-critical as well, the second one across all 7 of its deletion classes (Section 3.5). The smallest unit-margin obstruction we know of therefore remains at 19 vertices, and it is not for want of having looked one step below.

What the distinction is *not* visible to the predictability relaxation, which assigns h_2 's critical and free arcs alike no certificate at all, in line with Section 1.2.

3.5 Vertex-transitive and doubly regular tournaments

Beyond the Paley family we swept the structured families exhaustively up to 23 vertices. Every verdict is negative for the bound, so we report them briefly; what they buy is a map of where obstructions sit and where they do not.

First, **the 21-vertex vertex-transitive family contains no majority obstruction at all**: all 110 are 5-inducible, 106 of them already at unit margin and the remaining four by exhibited majority witnesses. Since only a majority refutation can move $N(5)$, that family is closed as a source of candidates below 23 — and the closure survives one-arc perturbation, since reversing any single arc of the four unit-margin obstructions, one representative per arc orbit, again leaves a majority-inducible tournament in all 40 cases. Second, *both* doubly regular tournaments on 19 vertices, Paley(19) and the one other of that order, are unit-margin obstructions and both are majority-inducible, so the separation that makes Paley(19) interesting is not a property of Paley(19) in particular. **Nor is its criticality**: the second one is *arc-critical* at unit margin too. Its automorphism group has order 3, so its 171 arcs fall into 57 orbits of size 3, and reversing the representative of any one of them yields a tournament with a unit-margin witness — 57 orbits, 57 witnesses, none of the searches stopped by a cap. Arc-criticality at unit margin is therefore shared by both doubly regular tournaments of order 19, which is the strongest form of criticality either could have and matches Paley(23)'s behaviour at unrestricted margin. Third, every regular tournament on at most 15 vertices is inducible at unit margin, over all $R_{15} \approx 1.8 \times 10^{10}$ of those on 15 vertices, with no instance left capped and with the instance counts of the 6,000 residues of *nauty*'s tournament generator [20] summing to OEIS A096368(7) [21] exactly. Fourth, every self-converse tournament on 13 vertices is inducible at unit margin, over all $S_{13} \approx 9.5 \times 10^7$ of them, with nothing capped; the sweep was split across the cluster and the laptop, and the listing of hosts is gated to exactly S_{13} lines before any of it is swept, so a truncated or duplicated catalogue stops the run rather than shrinking the claim. With the order-11 census of [2] and the order-12 analysis of Appendix C, that settles every self-converse tournament on at most 13 vertices. Across the structured families swept here, ten tournaments are unit-margin obstructions, six of them proved majority-inducible as well: two at 19 vertices, four at 21, and four at 23, of which only Paley(23) is settled at both margins. That count is a census of these families and not of every obstruction known — the arc reversals of Section 3.4 supply 210 further ones at 21 vertices, none of them vertex-transitive.

A refuted conjecture. Write $t(T)$ for the least number of cyclic triangles containing an arc of T . Every unit-margin obstruction we knew of had $t = 5$, and the one majority obstruction had $t = 6$, which suggested that a large t might be *necessary* for non-inducibility — an attractive possibility, since t is computable in $O(n^3)$ and would have given a cheap filter on candidates. At unit margin it is false: among the 23-vertex vertex-transitive tournaments with $t = 4$, three are obstructions while 36 others are inducible, so the triangle load neither implies nor precludes an obstruction. The unrestricted form of the conjecture survives, every majority obstruction we have satisfying $t \geq 6$; refuting it would require a majority obstruction with an arc lying in at most five cyclic triangles.

Costs. Every computation reported above, with its cost. A core-hour is the product of wall time and worker count. All were run on the machine described at the head of this section except the last two rows, which

include cluster time and so are not comparable with the others; Appendix A.5 records the factor between the two machines.

computation	margin	verdict	core-hours
Paley(19), certified refutation	unit	not inducible	25.1
Paley(23), certified refutation	unrestricted	not inducible	236.0
Paley(23), sweep	unrestricted	not inducible	34.03
Paley(27), sweep	unrestricted	not inducible	31.04
Paley(31), sweep	unrestricted	not inducible	26.89
Paley(43) – v , sweep	unrestricted	not inducible	185.5
Paley(23), one arc reversed	unrestricted	inducible	210.4
second doubly regular 19, all 57 arc reversals	unit	all inducible	3.59
the 15 obstruction reversals at 21, every vertex deletion	unit	all inducible	14.3
regular tournaments on 15 vertices	unit	all inducible	3,106
self-converse tournaments on 13 vertices	unit	all inducible	1,310

4. Discussion and open problems

4.1 Other applications of incremental depth-first search with MRV

Subject to the caveats of Appendix A.6, the shape of the engine may transfer: a placement search in which each decision *refines*, rather than recomputes, the candidate sets of all remaining objects, with a dynamic MRV order and an externally discharged symmetry break. The preconditions are specific - the extensions of a partial solution must factor as a product that shrinks monotonically under each decision, and the symmetry group must be large and computable — and where they fail we would expect a general-purpose solver to win. Problems that appear to share the shape include other tournament realisation questions and the construction of combinatorial designs with prescribed automorphisms, but we have not tested it outside the present setting.

Neither device is new in isolation, and the closest antecedent is Knuth’s Algorithm X with dancing links [22]. Its column-choice rule — take the column with the fewest remaining ones — is our MRV order, and its link surgery does what our refinement does: the state is altered in place and put back on backtrack, so a node costs what changed rather than what the whole state contains. The constraint is what differs. Exact cover partitions, whereas an arc of a tournament imposes a threshold, at least three of five voters, so the search here is not an exact cover instance and dancing links does not apply to it as it stands. What carries over is the pair of devices rather than the algorithm, and the reason it carries over is the product structure of Appendix A.3: the extensions of a partial profile factor as a choice of slot per voter, and it is the monotone refinement of that product, not a covering structure, that lets one empty factor prune a subtree.

Two design lessons do seem to generalise, being about how to build such a search rather than about when to prefer one. First, the incremental refinement, not the heuristic order, carried most of the speedup here, a factor of 110 against a factor of a few. Second, a symmetry break should be *parameterised* by its representatives rather than hard-coded, so that its correctness obligation becomes a separate computation instead of an assumption buried in the search; Section 3.3 is a case where discharging that obligation required an explicit orbit calculation that the engine could not have performed for itself.

4.2 Open problems

The value of $N(5)$. The bracket is now $13 \leq N(5) \leq 23$, with both ends constructive and both moved from [2]. Closing it requires either a non-inducible tournament on fewer than 23 vertices or an exhaustive argument at $n = 13, \dots, 22$. Note that only a *majority* refutation can move the upper end: the margin-1 obstructions below, at $n = 19$ and 21, bound nothing about $N(5)$, since every one of them is majority-inducible.

Where to look next. The state of the art for regular tournaments is that every one on at most 15 vertices is 5-inducible at unit margin (Section 3.5). Regularity is a strong hypothesis, and that bounds nothing about $n = 15$ in general. Since the regular ceiling on the number of voters required already exceeds the general one at $n = 11$, we would not expect the extremal instances to be regular, and the self-converse tournaments are the better next family.

Vertex- and arc-criticality. The state of the art is that Paley(23) is arc-critical unrestricted (Section 3.4). Whether Paley(43) minus *two* vertices is 5-inducible — equivalently whether Paley(43) minus one vertex is vertex-critical — is open, and within reach: $\text{Aut}(P_{43})$ is transitive on arcs and hence on pairs of vertices, so that tournament is unique up to isomorphism and one computation decides it.

The margin hierarchy. The first of the inclusions $\text{margin } 1 \subseteq \text{margin } 3 \subseteq \text{unrestricted}$ is known to be strict (Sections 3.2 and 3.5). Whether the second is ever strict — a tournament inducible by five voters but not at $\text{margin} \leq 3$ — is open. The 3-cycle bound of Section 3.3 shows that such a tournament must have an arc lying in *no* directed triangle, that is an arc $u \rightarrow v$ with $N^+(v) \cap N^-(u) = \emptyset$, which forces $d^+(u) \geq d^+(v) + 1$ and in particular rules out every regular tournament. The search should therefore be directed at lopsided tournaments, not at perturbations of regular ones.

5. Reproducibility

5.1 The repository

The code, data and verdict files are at <https://github.com/Leonardini/TournamentsBeyond5Voters>. Its `CLAIMS.md` carries one row per statement made here, naming the artifact that holds the verdict and the command that re-derives it, and it ends with an explicit list of what the package does not establish. A permanent archive of the commit corresponding to the accepted version will be deposited on acceptance and its DOI recorded here; readers of the preprint should take the commit hash from the version they are reading.

The repository contains the search engine as a single C translation unit, the SAT encoding and cube tooling in Python, the certification driver, the independent witness verifiers, the tournaments as bit strings, and a verdict record for every result reported here. Each verdict records the exact command line, the base and break used, the coverage audit, the cost, and — for the certified results — the two root hashes. An acceptance script rebuilds both published certificate roots from the per-cube hash chain and audits the coverage of every distributed sweep; each of its checks is paired with a control that must fail, so that a check which has stopped testing anything is visible as such.

5.2 What has been reproduced so far

Two independent implementations decide the same instances: the depth-first engine of Appendix A and the SAT pipeline of Appendix B.1. They share the two human lemmas of Appendix B.2 and no code. The margin-1 statements for Paley(19) and Paley(23) have six independent complete refutations across three different base decompositions; the unrestricted Paley(23) statement has two, from independent runs with different bases and different anchors.

Witness verification is likewise independent of the search: a separate script reads the tournament from its bit-string, reads the witness from the run log, and recomputes every arc’s support.

5.3 Which hashes must replicate and which may not

The certified computations publish two root hashes, and only one of them is portable.

ROOT (CNF) commits to the SHA-256 of every cube’s CNF, in cube order. These files are generated by the repository’s own code from the tournament and the base, and are reproducible on any machine: a third

party who regenerates the cube set *must* obtain this exact value. It is the meaningful cross-check, because it proves they solved the same problems.

ROOT (proofs) additionally commits to the LRAT proof bytes. These are **not** portable. CaDiCaL is deterministic run to run on a fixed binary — we verified this, three runs identical — but its heuristics use floating-point scoring, so a different version, architecture or optimisation level may search differently and emit a different, equally valid proof. A different solver differs entirely: Glucose emits 2.36 MB of DRAT where CaDiCaL emits 2.79 MB of LRAT for the same leaf. This hash therefore detects corruption or tampering *within* a run and reproduces only on an identical build. **A mismatch here is not evidence of an error.**

Search costs: nodes replicate, hours do not. The same three-way distinction applies to the uncertified sweeps. *Verdicts* replicate under any correct implementation — that is the claim the paper makes. *Node counts* replicate exactly, but only with five things held fixed, every one of which can differ silently between two people running what they believe is the same computation:

1. the same binary, or at least identical tie-breaking in the variable order and in the domain refinement — ties are common and a different rule explores a different tree of the same size class but not the same size;
2. the same vertex numbering of the tournament;
3. the same base size and base;
4. the same resulting set of base states, which follows from (2) and (3) but is worth checking directly, since it is the width of the outer loop;
5. the same margin regime.

The engine prints all of these except (2) in a header on every run — margin, variable order, MRV cap, tie-break mode, symmetry break, base size, base, base arc pattern and base-state count — so a run log is a self-describing fingerprint. Condition (2) is the tournament file itself, which is why the repository stores tournaments rather than regenerating them from their constructions: one rebuilt from its Cayley presentation need not carry the numbering we used, and two of the four 21-vertex tournaments of Section 3.4 are in fact *not* in the canonical labelling that a fresh isomorphism-class enumeration produces. *Wall-clock and core-hours* replicate only in order of magnitude; Appendix A.5 measures a factor of 2.6 to 3.3 per core between the two machines we used, though the self-converse census of Section 3.5 ran at 5.1, so that ratio is workload-dependent rather than a property of the pair of machines, and a loaded machine distorts them further.

One practical note for anyone rerunning the checker: `lrat-trim` signals success with `s VERIFIED` and exit code 20, following the SAT-solver convention, not exit code 0.

Motivation and statement of AI use

This paper grew out of the companion work [2], whose explicit bound $N(5) \leq 43$ came from a structural argument that does not descend, and whose counting bound $N(5) \leq 38$ names no tournament at all. The question of what a genuine decision procedure would reach, at the orders the screen cannot speak about, is what led to the engine of Appendix A.

During the preparation of this work, the authors used Claude for exploratory reasoning, implementation assistance, drafting and editing. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

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References

- [1] G. Bachmeier, F. Brandt, C. Geist, P. Harrenstein, K. Kardel, D. Peters, H. G. Seedig. k -Majority digraphs and the hardness of voting with a constant number of voters. *Journal of Computer and System Sciences*, 105:130–157, 2019.
- [2] L. Chindelevitch, A. Harutyunyan. Tournaments determined by three and five voters. *Submitted*.
- [3] M. J. H. Heule, O. Kullmann, V. W. Marek. Solving and verifying the Boolean Pythagorean triples problem via cube-and-conquer. In *Theory and Applications of Satisfiability Testing (SAT 2016)*, LNCS 9710, pages 228–245, 2016. arXiv:1605.00723.
- [4] N. Alon, G. Brightwell, H. A. Kierstead, A. V. Kostochka, P. Winkler. Dominating sets in k -majority tournaments. *Journal of Combinatorial Theory, Series B*, 96(3):374–387, 2006.
- [5] R. L. Graham, J. H. Spencer. A constructive solution to a tournament problem. *Canadian Mathematical Bulletin*, 14(1):45–48, 1971.
- [6] D. Fidler. A recurrence for bounds on dominating sets in k -majority tournaments. *Electronic Journal of Combinatorics*, 18(1):P166, 2011.
- [7] C. Eggermont, C. Hurkens, G. J. Woeginger. Realizing small tournaments through few permutations. *Acta Cybernetica*, 21(2):267–271, 2013.
- [8] D. Shepardson, C. A. Tovey. Smallest tournaments not realizable by $\frac{2}{3}$ -majority voting. *Social Choice and Welfare*, 33(3):495–503, 2009. doi:10.1007/s00355-009-0375-7.
- [9] B. Dushnik, E. W. Miller. Partially ordered sets. *American Journal of Mathematics*, 63(3):600–610, 1941.
- [10] M. J. H. Heule, O. Kullmann, S. Wieringa, A. Biere. Cube and conquer: guiding CDCL SAT solvers by lookaheads. In *Haifa Verification Conference (HVC 2011)*, LNCS 7261, pages 50–65, 2012.
- [11] L. Cruz-Filipe, M. J. H. Heule, W. A. Hunt Jr., M. Kaufmann, P. Schneider-Kamp. Efficient certified RAT verification. In *Automated Deduction (CADE 2017)*, LNCS 10395, pages 220–236, 2017. This is the paper introducing the LRAT format that `lrat-trim` consumes.
- [12] L. Cruz-Filipe, J. Marques-Silva, P. Schneider-Kamp. Formally verifying the solution to the Boolean Pythagorean triples problem. *Journal of Automated Reasoning*, 63(3):695–722, 2019. doi:10.1007/s10817-018-9490-4.
- [13] M. J. H. Heule. Everything’s bigger in Texas: the largest math proof ever. ACL2 Seminar, University of Texas at Austin, 9 September 2016. Joint work with O. Kullmann and V. W. Marek. <https://www.cs.utexas.edu/~moore/ac12/seminar/2016.09.09-heule/ACL2.pdf> Cited for the compressed size of the published cube certificate; the re-verification costs quoted in Appendix B.1 are stated in [3] itself.
- [14] M. J. H. Heule. Schur number five. In *AAAI 2018*, pages 6598–6606, 2018.
- [15] J. Brakensiek, M. Heule, J. Mackey, D. Narváez. The resolution of Keller’s conjecture. In *Automated Reasoning (IJCAR 2020)*, LNCS 12166, pages 48–65, 2020.
- [16] D. C. McGarvey. A theorem on the construction of voting paradoxes. *Econometrica*, 21(4):608–610, 1953.

- [17] R. Milosz, S. Hamel, A. Pierrot. Median of 3 permutations, 3-cycles and 3-hitting set problem. In *Combinatorial Algorithms (IWOCA 2018)*, LNCS 10979, pages 224–236, 2018.
- [18] A. Biere, K. Fazekas, M. Fleury, M. Heisinger. CaDiCaL, Kissat, Paracooba, Plingeling and Treengeling entering the SAT Competition 2020. In *Proceedings of SAT Competition 2020: Solver and Benchmark Descriptions*, volume B-2020-1 of *Department of Computer Science Series of Publications B*, pages 50–53. University of Helsinki, 2020. The solver description; for the LRAT support we use, and for `lrat-trim`, see [24].
- [19] M. J. H. Heule, W. A. Hunt Jr., N. Wetzler. Trimming while checking clausal proofs. In *Formal Methods in Computer-Aided Design (FMCAD 2013)*, pages 181–188, 2013. The `drup-trim` checker, predecessor of `drat-trim`; the tool itself is [23].
- [20] B. D. McKay, A. Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. The `nauty` and `gentourng` tools, used for all tournament enumeration.
- [21] OEIS Foundation Inc. Entry A096368, *Number of unlabeled regular tournaments with $2n + 1$ nodes*. The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A096368>.
- [22] D. E. Knuth. Dancing links. In J. Davies, B. Roscoe, J. Woodcock, editors, *Millennial Perspectives in Computer Science*, pages 187–214. Palgrave, 2000. arXiv:cs/0011047.
- [23] N. Wetzler, M. J. H. Heule, W. A. Hunt Jr. DRAT-trim: efficient checking and trimming using expressive clausal proofs. In *Theory and Applications of Satisfiability Testing (SAT 2014)*, LNCS 8561, pages 422–429, 2014. doi:10.1007/978-3-319-09284-3_31.
- [24] F. Pollitt, M. Fleury, A. Biere. Faster LRAT checking than solving with CaDiCaL. In *Theory and Applications of Satisfiability Testing (SAT 2023)*, LIPIcs 271, pages 21:1–21:12, 2023. doi:10.4230/LIPIcs.SAT.2023.21. Introduces `lrat-trim` and CaDiCaL’s native LRAT output, which is the capability we use; it postdates the solver description [18].
- [25] OEIS Foundation Inc. Entry A000568, *Number of outcomes of unlabeled n -team round-robin tournaments*. The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A000568>.
- [26] OEIS Foundation Inc. Entry A002785, *Number of self-complementary oriented graphs with n nodes*. The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A002785>. McKay’s note on that entry records that these are also the self-converse tournaments.

Appendix A. The search engine

Both integer programming and satisfiability are central to this project rather than foils for it. Every computational claim in the companion paper [2] is certified by an integer or linear program, and the integer program of A.1 below is that paper’s formulation B.1, verbatim; the only machine-checked theorem in this paper is a SAT computation from end to end, with CaDiCaL [18] emitting LRAT proofs that `lrat-trim` checks. The claim here is narrow: on this one decision problem, at this one range of sizes, neither tool reaches the answer while a search built around the problem’s own structure does — and the same SAT machinery is then indispensable for turning that answer into a proof.

A.1 Integer programming is too slow

The natural integer program has one binary $x_{v,ij}$ per voter v and pair $i < j$, meaning that v ranks i above j , with $x_{v,ji}$ read as $1 - x_{v,ij}$ so that antisymmetry is structural. Each voter is forced to be a linear order by forbidding both orientations of every 3-cycle,

$$0 \leq x_{v,ij} + x_{v,jl} - x_{v,il} \leq 1 \quad (i < j < l),$$

and the profile is forced to induce T by one majority row per arc, $\sum_v x_{v,ij} \geq (k+1)/2$. There is no objective: this is a pure feasibility problem. At $q = 23$, $k = 5$ it has 1,265 binaries, 17,710 transitivity rows and 253 majority rows.

The obstruction is structural rather than a deficiency of the solver. Put $x_{v,ij} = 3/5$ on every arc of T and $2/5$ on every non-arc, the same for all voters. Every cyclic triple then reads $3/5 + 3/5 - 2/5 = 4/5$ and every transitive one $3/5$, both interior to $[0, 1]$, while every majority row is tight at exactly 3. The relaxation is therefore feasible, and since the problem carries no objective, LP feasibility yields no bound whatever: branch-and-bound is left with nothing to prune by except infeasibility discovered deep in the tree. That fractional point is also invariant under both symmetry groups acting on the problem — the $k!$ voter relabellings and $\text{Aut}(T)$ — so it supplies no branching direction either, and orbital branching recovers only the $k!$ factor. CPLEX and Gurobi both failed to close instances at $n = 19$ within a day.

A.2 SAT is too slow

The dissent-Boolean encoding of Section 1.1 is compact and propagates well locally. It is the right encoding — we use it ourselves for the certified result — but as a *decision procedure* on a symmetric instance it is dominated by the search below.

Table 4 of [1] reports their **Sat-Check-k-Majority** on uniform random tournaments, tabulating only entries whose average stays under 20 seconds. The $k = 5$ column reads

n	18	19	20	21	22	23	24
seconds	0.23	0.35	0.54	5.87	11.07	18.95	—

so on *random* instances the method leaves the table between $n = 23$ and $n = 24$, roughly doubling per added vertex. Random tournaments are the easy case: no automorphisms to rediscover and, being far from extremal, many witnesses. On the structured instance of the same order, Paley(23), the same solver ran for six weeks without terminating. The gap between 18.95 seconds and six weeks at equal n is the cost of symmetry.

A better encoding and an explicit symmetry break both help substantially and are still not enough. The algorithm of [1] encodes each voter by order variables over kn^2 literals and expresses the majority condition through a Tseitin transformation; the dissent-Boolean encoding carries one variable per (arc, voter) pair, $k\binom{n}{2}$ in all, and needs no auxiliaries, the majority condition being exactly the $\binom{5}{3} = 10$ clauses forbidding each 3-subset of dissenters. On doubly regular test tournaments, CaDiCaL conflict counts were 2,399 for the order-variable family, 1,491 for an intermediate pair-label encoding and 57 for the dissent-Boolean one, a 26 to 42-fold reduction. We then implemented three breaks: a lex chain on the voters’ dissent vectors, a strengthening forbidding repeated voters, and an Aut-based pin fixing a cyclic triangle. (The last two must not be combined with the first, for the reason in Appendix A.3: a pin and a lex chain can spend the same group twice.) With these active, neither Paley(19) nor the second doubly regular tournament of order 19 returned a verdict.

So a 40-fold better encoding and a sound symmetry break together leave the instance out of reach. What does work is supplying the solver with the decomposition itself — cube-and-conquer over base states, which is how both refutations are certified in Appendix B. A CDCL solver must otherwise discover that decomposition for itself, and cannot exploit $\text{Aut}(T)$ at all without an explicit break in the encoding. SAT earns its place in this paper as a *proof format* rather than as a decision procedure; and once one has the decomposition, one may as well run it directly, which is the search below.

A.3 Depth-first placement search

The search maintains a partial profile: a set $S \subseteq V$ of *placed* vertices together with, for each voter, a linear order of S . It begins from a base state, which places B , and extends by inserting one unplaced vertex at

a time. Inserting v means choosing, for each voter independently, a slot among the $|S| + 1$ available; the domain $D(v \mid S)$ is the set of k -tuples of slots consistent with every arc between v and S . A partial profile is extendable only if every unplaced vertex has a non-empty domain, and domains shrink monotonically as S grows, which makes the search a constraint-propagation problem rather than a blind enumeration.

ALGORITHM 1. Placement search for k -inducibility of T

Input : tournament T on vertex set V ; number of voters k ;
margin regime (majority: support $\geq (k+1)/2$; unit: support $= (k+1)/2$)
Output: a witness profile, or a refutation

MAIN

```

for each base state (a lex-canonical assignment of the base  $B$  to  $k$  orders):
   $S \leftarrow B$ 
   $D(v \mid S) \leftarrow$  fresh enumeration, for every  $v$  not in  $S$ 
  if any  $D(v \mid S)$  is empty: continue // this base state is dead
  if DFS( $S$ ) succeeds: return its witness
return "not  $k$ -inducible" // no base state extends

```

DFS(S)

```

if  $S = V$ : return the profile // all vertices placed
 $v^* \leftarrow$  argmin  $|D(v \mid S)|$  over  $v$  not in  $S$  // MRV, recomputed at EVERY node
ties broken toward max  $|out(v) \cap S| - |in(v) \cap S|$ 
for each slot tuple  $p$  in  $D(v^* \mid S)$ :
  insert  $v^*$  at  $p$  in each of the  $k$  orders;  $S' \leftarrow S \cup \{v^*\}$ 
  for each  $w$  not in  $S'$ :
     $D(w \mid S') \leftarrow$  REFINE( $D(w \mid S)$ ,  $p$ ,  $w$ ,  $v^*$ )
    if  $D(w \mid S')$  is empty: undo; continue with the next  $p$ 
  if DFS( $S'$ ) succeeds: return its witness
  undo // arena pointer reset
return failure

```

```

REFINE( $D$ ,  $q$ ,  $w$ ,  $u$ ) //  $q$  = the slot tuple just used for  $u$ 
 $D' \leftarrow \{\}$ 
for each  $p$  in  $D$ :
  lift  $p$  to  $p'$  voter by voter:
     $p_i < q_i \rightarrow p'_i = p_i$  //  $w$  stays before  $u$ 
     $p_i > q_i \rightarrow p'_i = p_i + 1$  //  $w$  stays after  $u$ , shifted by the insertion
     $p_i = q_i \rightarrow$  BOTH are legal, so the tuple splits in two
  keep those lifts whose support  $c_w(u) = \#\{i : w \text{ before } u\}$ 
  meets the requirement of the single new arc between  $w$  and  $u$ 
return  $D'$ 

```

Four choices make it fast, and the second is the one that made $n = 19$ decidable at all.

A dynamic variable order. At every node we insert the unplaced vertex minimising $|D(v \mid S)|$, breaking ties toward the vertex whose arcs to S are most imbalanced. The important part is that the order is recomputed at every node rather than fixed per base state: on Paley instances a static order is not competitive, because how constrained a vertex is depends strongly on which part of the tournament has already been placed.

Incremental domain refinement. Recomputing $D(v \mid S)$ from scratch at each node is the dominant cost of a naive implementation; instead the child's domain is derived from the parent's by the three cases in REFINE above, after which the single new arc filters the lifted tuples by their support. Child domains are therefore refinements of parent domains, computed in time proportional to the surviving tuples rather than

to the whole slot space, and domains live in a stack-disciplined arena indexed by (depth, vertex) so that backtracking is a pointer reset. Like for like this is worth a factor of about 110 in wall time.

Only part of that factor is speed per node, and the rest is a point worth separating. Before incremental maintenance, computing $|D(v \mid S)|$ meant a fresh enumeration, so the implementation counted each domain only up to a cap. A cap suffices to detect the empty domain — which is what prunes — but it destroys the ordering among everything above it, and the argmin degenerates to an arbitrary choice. On a 29-instance benchmark drawn from Paley(19), with identical verdicts throughout, the search visited 992,378 nodes under the capped rule and 294,964 under the exact one: a factor of 3.4 in tree size that is pure decision quality, leaving roughly 32 of the 110 as cost per node. The domain statistics explain why the cap hurt so much here — the mean $|D(v \mid S)|$ on Paley(19) is 64 at $|S| = 5$, falling to 8.2 at $|S| = 11$, where most of the search lives, and 4.8 at $|S| = 12$ — so a cap of five sat *below* the mean almost everywhere the search spent its time. Raising the cap is no fix on its own, since under fresh enumeration a higher cap costs proportionally more at every node; that is what made the capped design right *before* the refinement existed. Exact selection was not worth having until it became free.

Symmetry breaking. The $k!$ voter symmetry is absorbed exactly by the base-state definition, which counts non-decreasing tuples only. For $\text{Aut}(T)$ the engine imposes the anchoring of Lemma 2.1 or of Corollary 2.2, whichever is stronger on the tournament: the condition is invariant under permuting voters, so it composes with the base lex-order, and monotone under insertion, so it can be tested during the search rather than only at the leaves. Its soundness obligation is external — the engine cannot compute $\text{Aut}(T)$ and so cannot check that the supplied representatives meet every orbit — and Section 2 sets out how we discharge it.

Early termination and caching by arc pattern. A cheap per-voter agreement bound, obtained by counting the agreement between voter i and the arcs between v and S two ways, proves $D(v \mid S)$ empty in $O(k|S|)$ time without enumerating it. And the per-base-state work is keyed by the *arc pattern* of $T|_B$ rather than by B itself, so base-state counts — which depend only on that pattern — are computed once and reused across every copy of the same base class and across every q .

A.4 Choosing the decomposition

The base set is not a free parameter: different bases give wildly different total cost, and the ranking is not by base-state count. For each of the twelve isomorphism classes of 5-vertex tournament we compute, without search, the number of base states and the number surviving the automorphism break, then measure seconds per base state on a sample and rank configurations by the *product*. This matters: in our Paley(31) experiments the class with 4,007 surviving base states beat the class with 2,537 on total cost, because its states are individually much cheaper, so ranking by count alone would have chosen the worse configuration.

A.5 How the cost scales in q

On a fixed base decomposition the total cost **decreases** as the tournament grows. Measured on 24 bit-identical base states of the same 5-vertex base class, on one core with the other nine busy:

ratio	total time	nodes	μ s per node
$q = 31$ over $q = 27$	0.933	0.614	1.521
$q = 43$ over $q = 31$	0.746	0.385	1.935
$q = 43$ over $q = 27$	0.739	0.237	2.94

Two steep opposing trends, not a plateau: node counts fall roughly like n^{-3} , because a larger Paley tournament is *more* constrained and refutation fires higher in the tree, while per-node work rises roughly like $n^{+2.3}$, and the fall keeps winning. It can win only because the outer loop is independent of q : a 5-vertex base class has the same number of base states at every q , that being a property of the base pattern and not of the

tournament, so growing the tournament adds work per node without widening the decomposition. The full runs bear this out:

q	core-hours	nodes	seconds per live base state
23	34.03	—	47.5
27	31.04	1.14×10^8	44.0
43	22.87	1.43×10^9	32.5

Paley(43) is decided in **less time than Paley(27)**, despite exploring an order of magnitude more nodes, and that run is an independent re-derivation of the main theorem of [2] by a method sharing nothing with the co-backing screen. The scope of the claim is exactly those three values of q on this one decomposition, where growing q tightens the constraints. It is not a claim that every larger instance is cheaper — Paley(43) minus a vertex costs 185.5 core-hours, far more than Paley(43) itself, because deleting a vertex *removes* constraints and weakens the automorphism group at once, and it averages 83.2 seconds per base state with a maximum of 313.2 — nor that the trend continues past the range measured.

A second, independent gain runs the same way: the search gets much cheaper as the margin is tightened. On the identical tournament, engine and 5-vertex base class,

Paley(43) – v , $n = 42$	base states	core-hours
margin ≤ 1	2,200	2.07
unrestricted	8,031	185.5

a factor of about 90. Part is the outer loop — requiring every arc to sit at exactly 3–2 admits 2,200 base states where majority admits 8,031 — and the rest is per state: under exact margin the last voter is *forced*, since the support accumulated over the first $k - 1$ voters determines what the last must contribute, so a whole level of the search collapses. The pattern holds across the family and again improves with n : complete margin-1 refutations cost 2.86 core-hours at $n = 26$, 2.21 at $n = 30$ and 2.07 at $n = 42$. We do *not* claim the converse for the general-purpose methods: tightening the margin adds constraints to both encodings, which usually makes a refutation easier, so the effect there could run either way and we have not measured it. Our claim is one-sided and about our own engine — faster on the more constrained question and faster on the larger tournament, which is the opposite of what one expects from a search procedure, and is what makes the approach usable at these sizes.

Two machines, and why their costs are not comparable. Some computations ran on a cluster rather than the laptop. Measured across three independent campaigns, the cluster ran at 2.6 to 3.3 times the laptop’s cost per unit of work, the larger machine being *slower* per core, so a core-hour figure means nothing here without the machine that produced it.

A.6 The scope of the comparison

We are not claiming that bespoke search beats integer programming or satisfiability in general — it does not, and the reasons general-purpose solvers dominate most of combinatorial optimisation still apply. This engine is narrow: it exploits the specific facts that a partial profile’s extensions form a product of per-voter slot choices, that this product refines monotonically, and that $\text{Aut}(T)$ is large and computable, and removing any one of those leaves it with no advantage. The claim is bounded three ways at once — it concerns k -inducibility of highly symmetric tournaments at $k = 5$ for $19 \leq n \leq 43$; it reflects *our* implementations, so a better-tuned model or a cleverer CNF might change the numbers; and it uses the best formulations we could construct for each method, which for SAT means an encoding measured 26 to 42 times better than the published one and for the integer program means formulation B.1 of [2] rather than a straw man. Within those bounds the comparison is real: instances the general-purpose methods did not settle for us, the search settles in hours.

Appendix B. The formal certificates

B.1 The certified refutations

Both machine-checked results follow an established design rather than a new one. *Cube-and-conquer* is due to Heule, Kullmann, Wieringa and Biere [10]: split a hard instance into many partial assignments (*cubes*), hand each to a CDCL solver, and let the splitting supply the global structure the solver cannot find for itself. The standard of *proof* is the one Heule, Kullmann and Marek set for the Boolean Pythagorean triples problem [3], and Schur number five [14] and the Keller conjecture [15] after it: every subproblem emits a machine-checkable refutation, validated by an independent checker rather than trusted. We claim no methodological novelty.

Two departures are worth stating, both consequences of our setting. Our cubes are not produced by lookahead splitting but are *base states*, a combinatorial decomposition we have independently from Appendix A — which is what lets the coverage of the cube set be certified separately (B.2) instead of resting on a splitting heuristic. And where [3] published a cube certificate from which its refutations can be regenerated, we *verify and discard*: each LRAT proof is checked, hashed and deleted, so peak storage is one proof per worker rather than terabytes, and what we publish is a hash commitment. That is the weaker artifact — a reader cannot re-check our proofs without regenerating them — and Section 5.3 says which published hashes a third party must reproduce. The choice suits the scale. Re-verifying the Boolean Pythagorean triples proof costs about 13,000 CPU-hours to regenerate the refutations from the published cube certificate, 68 gigabytes compressed [13], and about 16,000 to check them [3], which is why that proof ships as a certificate to re-solve rather than as its 200 terabytes of proof bytes. Our certificates are two to three orders of magnitude smaller: 25.1 core-hours for Paley(19) and 236.0 for Paley(23) cover solving *and* checking from nothing, so a reader who distrusts our hashes can regenerate an entire computation for between a thousandth and a hundredth of what validating that proof costs, and compare portable roots — which is exactly the check Section 5.3 supports.

For Paley(19) at margin 1 the instance is the dissent-Boolean encoding of Section 1.1 at $q = 19$, $k = 5$: 855 variables and 12,265 clauses. The cubes are the base states of the 6-vertex base $\{1, 2, 3, 4, 5, 12\}$ — 142,251 of them, of which 22,876 survive the arc-orbit break — and each is handed whole to CaDiCaL with `--lrat=true --checkproof=0`. The self-check is *deliberately disabled*: the point is that a different program validates the proof, so the solver must never be permitted to validate its own work. Each LRAT proof is checked by `lrat-trim`, then hashed and discarded. The cost was 24.3 core-hours to solve and 0.8 to check, 2.64 hours wall on ten workers. We chose LRAT over DRAT on measurement: on real leaves from this instance, Glucose with DRAT plus `drat-trim` [23], which trims while it checks [19], cost 1.043 s per leaf against CaDiCaL with LRAT plus `lrat-trim` at 0.295 s — a factor of 3.5 end to end and of 29 in checking alone, because LRAT carries explicit clause hints and the checker performs no backward search. Because the cubes are base states, no *deepening* step is involved and no uncertified enumeration enters the trust chain.

The same two-part certificate has since been completed for the Paley(23) refutation of Section 3.1, at margin *unrestricted*, so that theorem is machine-checked on both halves. Its parameters are the base $\{1, 2, 3, 4, 6, 7\}$ with 3,414,729 base states, of which 343,896 survive the arc-orbit and non-arc-orbit breaks and become live cubes; every one is UNSAT with an LRAT proof independently verified by `lrat-trim`, at 228.3 core-hours to solve and 7.7 to check, 21.72 hours wall on eleven workers, with the cube split adding 0.39 hours for the 22.11-hour end-to-end figure quoted in the abstract. The coverage half is a single unsatisfiable instance of 3,415,435 clauses — 350 constraint clauses, 356 symmetry-breaking clauses and 3,414,729 negated cubes, one per base state, since coverage must see *every* base state whereas the orbit breaks apply to the search half only, which is why it is ten times the live cube count. It solves in 1,182.95 seconds and emits a 2,097 MiB LRAT proof, verified by `lrat-trim`, with peak resident memory 5.6 GB; reaching that required streaming the clauses to disk rather than building the instance in memory, two earlier in-memory attempts having been killed at 8.99 GB.

B.2 What the certificates do and do not establish

The trust chain has four links:

link	status
every live cube is unsatisfiable	machine: CaDiCaL, rechecked by <code>lrat-trim</code>
the cubes cover every lex-canonical base assignment	machine: DRAT, verified by <code>drat-trim</code> [23]
(L1) orbit anchoring loses no generality	human (Lemma 2.1)
(L2) lex-ordering the voters loses no generality	human (Lemma 2.3)

The two human links are proved in Section 2. Lemma 2.1 is what licenses solving only the *live* cubes, in its arc/non-arc form: $\text{Aut}(P_q)$ is transitive on arcs and on non-arcs, so a voter’s top pair may be assumed to be a fixed arc or a fixed non-arc. Lemma 2.3 licenses the lex chain in the coverage computation. They are human links not because they are doubtful but because nothing in the pipeline checks them; the depth-first engine inherits the same two, shares no implementation with this one, and where the two agree the agreement is evidence about the implementations rather than about the lemmas.

One episode is worth recording, because for a period this certificate asserted more than it had established. The script that writes the verdict emitted “coverage of the cube set: certified separately” *unconditionally*, without running the coverage check, which is a different program. The claim was true of Paley(19), where coverage had been run, and false of Paley(23) at the time it was printed. It now reports coverage status rather than asserting it. A certificate that describes its own scope incorrectly is the failure mode this section exists to guard against, and no check caught it: it was caught by re-reading the verdict against the log of what had actually run.

The combined Paley(23) certificate, binding split and search into one value, is `ff60539e16abf2ec...`, with the search root at `7e6c9c26ac386e67...` and the coverage instance at `4876f077e85b7b56...`; Section 5.3 says which a third party must reproduce. Total cost on one laptop was 22.11 hours: an instance a general-purpose solver did not finish in six weeks [1] is settled, with an independently rechecked proof of every subproblem, in under a day on commodity hardware.

Appendix C. The order-12 case analysis

Every tournament on 12 vertices is 5-inducible, so $N(5) \geq 13$. This appendix describes that computation. It is the one result in this paper that was not run on the laptop of Section 3: it ran on a shared cluster, and its cost is reported in cluster core-hours, which Appendix A.5 records as 2.6 to 3.3 times more per unit of work than this laptop’s on the computations measured here; the self-converse census of Section 3.5 ran at 5.1, so we report the range rather than a single factor and size new work by the worst of them.

C.1 Reduction to one-vertex extensions

A direct census is out of reach: $D_{12} \approx 1.5 \times 10^{11}$ classes, 170 times the order-11 census of [2]. We use that census as the outer loop instead.

Write $L + S$ for the tournament obtained from an order-11 tournament L by adding a vertex v whose out-neighbourhood is $S \subseteq V(L)$. Two observations organise the analysis.

(C1) Every order-12 tournament is $L + S$ for some order-11 L and some S — indeed for each of its twelve one-vertex deletions.

(C2) Restricting a profile that realises $L + S$ to $V(L)$ gives a profile that realises L .

By (C1), enumerating all D_{11} order-11 classes and settling all $2^{11} = 2,048$ extensions of each settles every order-12 tournament; all of the former are 5-inducible by the census of [2], which is what makes the outer loop available. The twelvefold redundancy is harmless: this is a case analysis, not a minimum cover, and each class need only be settled once. By (C2), a profile realising $L + S$ may be sought by *inserting* v into a profile realising L with no loss of generality, since every profile for $L + S$ arises that way. That is what phase 1 exploits.

C.2 Three phases

The phases differ in what they can conclude, and are ordered by cost.

Phase 1, the covering search. Rather than 2,048 separate instances per order-11 tournament, one search enumerates profiles realising L and, for each, records every out-neighbourhood obtainable by inserting v into its five orders: v 's position in each order is a threshold, and the support of $v \rightarrow u$ is the number of orders that place v above u , so one traversal certifies many extensions at once. This is the engine of Appendix A.3 with v 's placement read off rather than fixed in advance. The phase is one-sided and limited by a work allowance — an extension it covers is thereby certified 5-inducible, one left over is not thereby refuted — and it settles every extension of *[PLACEHOLDER — final share settled at phase 1; 99.32% at 41.6% of the census]* of the order-11 tournaments.

Phase 2, deciding what is left. Each extension phase 1 leaves is materialised as an order-12 tournament $L + S$ in its own right and decided by the same engine at unit margin, where a witness settles it positively. *[PLACEHOLDER — final count reaching phase 2; 2,257,151 so far]* extensions reached this phase, and every one of them is 5-inducible with unit margin.

Phase 3, plain majority. Only a *majority* refutation would be a counterexample, so any instance negative at unit margin is re-run without the margin restriction. **Phase 3 was never invoked:** nothing was negative at unit margin, so the weaker question never had to be asked.

The division of labour between the phases is not a tuning choice. An earlier design gave a stubborn L more covering time instead of deciding the remainder directly, and on the structured tournaments that sit at the top of the generator's enumeration tree it terminated at no limit we tried: those leave hundreds of extensions uncovered, and no amount of further traversal covers an extension that no insertion into any profile for L can reach. Deciding it as an instance of its own is what makes the analysis finite.

C.3 One tournament of each converse pair

If $\pi_1, \dots, \pi_5$ realise T , their reversals realise the converse $\bar{T}$, arc for arc and with the same margins, so T is 5-inducible if and only if $\bar{T}$ is. Moreover $\overline{L + S} = \bar{L} + (V(L) \setminus S)$, so settling all 2,048 extensions of L settles all 2,048 of $\bar{L}$ as well. We therefore keep one tournament of each converse pair, by the rule that L is kept if and only if $\text{canon}(L) \leq \text{canon}(\bar{L})$. This keeps exactly one of each pair together with every self-converse tournament, $(D_{11} + S_{11})/2$ in all, where S_{11} is the number of self-converse order-11 classes; the saving is a factor just under two.

Both halves of that argument are tested against enumeration at orders 5 to 8, where the truth is small enough to compute directly: the rule keeps 10, 34, 272 and 3,528 of the 12, 56, 456 and 6,880 classes, matching $(D_n + S_n)/2$ for $S_n = 8, 12, 88, 176$; and the identity $\overline{L + S} = \bar{L} + (V(L) \setminus S)$ is checked on all 384, 3,584 and 58,368 pairs (L, S) at orders 5, 6 and 7.

C.4 The completeness gate

A count of zero counterexamples is equally consistent with a complete sweep and with a sweep that examined nothing, so it is not the quantity to check. The gate is the census: the number of order-11 classes generated, summed over the parts, must equal D_{11} exactly. Each part records itself complete only on success, so one that is interrupted or that runs out of its allowance simply remains outstanding and is redone, and the set of recorded parts is the certificate; the aggregate reports no verdict until the census total is met. It also checks the number of tournaments kept against $(D_{11} + S_{11})/2$ — a quantity no part can verify alone, since the two members of a converse pair generally fall in different parts.

We are explicit about this because an earlier run of this pipeline reported every part complete, no counterexamples and no failures, while examining zero tournaments: the generator was not present on the compute nodes, its error output was discarded, and a part that produced nothing was recorded as a part that had

found nothing. No line of that report was false. It was simply consistent with both outcomes, which is the same failure of scope as the certificate episode of Appendix B.2, one layer up.

Total cost: *[PLACEHOLDER — final figure; 4,015 cluster core-hours at 41.6% of the census]*. The search code, the per-part records and the roll-up are at `cluster/jz_n12cover` in the repository of Section 5.

Appendix D. Exact enumeration counts

The counts the main text gives to two significant figures, in full. D_n , R_n and S_n are the numbers of isomorphism classes of all, of regular, and of self-converse tournaments on n vertices. The values at $n \leq 8$ are the ones the arguments of Appendix C.3 are tested against, and D_{11} is the completeness gate of Appendix C.4.

n	D_n	R_n	S_n
5	12	1	8
6	56	—	12
7	456	3	88
8	6,880	—	176
9	191,536	15	2,752
10	9,733,056	—	8,784
11	903,753,248	1,223	279,968
12	154,108,311,168	—	1,492,288
13	—	1,495,297	95,458,560
15	—	18,400,989,629	—

A dash marks a value this paper does not use; [2] follows the same convention and collects its own censuses in its Appendix H, with which this table agrees where the two overlap. The D_n column is OEIS A000568 [25] and the S_n column is OEIS A002785 [26]; we checked the latter against our own exhaustive count at $n = 10$, which found 8,784 self-converse classes among all 9,733,056, and its $n = 12$ term is the size of the family Appendix C sweeps. Note that [2] reads R_n as *semi*-regular in even order, and that we need it in odd order only. The R_n column is OEIS A096368 [21], which is indexed so that its term m counts the regular tournaments on $2m + 1$ vertices; the value for 13 vertices is therefore the term $m = 6$, and that is the one our own exhaustive run reproduces, which is what anchors the indexing.

The converse-halving check of Appendix C.4 compares the number of tournaments kept against $(D_{11} + S_{11})/2 = 452,016,608$, which the table makes exact. That the two counts have the same parity is itself a necessary condition, and it holds. The check exists to catch a keep rule that halves nothing or halves twice, and an exact target catches more than that: it also catches a rule that halves correctly on all but a handful of converse pairs, which the factor 1.999 would not distinguish from correct.

For the record, the explicit tournament of [1] mentioned in Section 1 has 603,979,799 vertices.

No figures. The one worth drawing is a worked example of the search on Paley(7) at $k = 3$, small enough to print in full, and it would earn its space only if a referee finds Appendix A.3 hard to follow. The other planned appendices — base-class tables, verdict files verbatim, the Paley(19) witness with its support histogram — are in the repository of Section 5 instead.